

Principal Component Analysis

Lab 3

Instructions

This lab assignment exposes you to working with Principal Component Analysis (PCA) to expose how they can determine the best explained variance ratio. This is a statistical procedure that is used to reduce dimensionality. It makes the data visualizations easy to handle. PCA also decreases the complexity of the model and increases computational efficiency before starting with predictive analytics. There are three main steps:

1. Access Google Colab by signing into Google
2. Create a new notebook and install and import Python Packages
3. Download dataset and summarize data

Lab Background

For the lab, we will be using the dataset “b5.csv”, a dummy data set with 18930 rows and 50 columns variables. PCA is most useful when data features are highly correlated. There are six steps involved in PCA.

- Step 1: Standardize the dataset.
- Step 2: Calculate the covariance matrix for the features in the dataset.
- Step 3: Calculate the eigenvalues and eigenvectors for the covariance matrix.
- Step 4: Sort eigenvalues and their corresponding eigenvectors.
- Step 5: Pick k eigenvalues and form a matrix of eigenvectors.
- Step 6: Transform the original matrix

Step One:

Access Google Colab at: <https://colab.research.google.com/>. Select the blue ‘Sign in’ button from the top right corner of the page to login.

Note: *You will need a google account to be able to utilize the cloud platform.*

Step Two:

In Google Colab, select *File* from the top navigation bar and then *new notebook* to be able to install the Python Packages required for the lab.

- Rename your notebook as YourLastName_FirstName_LAB# (example: Daniels_Jerome_LAB1).

- Save a copy in your Drive by selecting 'File' and then 'Save a copy in Drive.'

Python Packages

```
#-----import all modules-----  
import matplotlib.pyplot as plt  
import pandas as pd  
import numpy as np  
import sklearn  
from sklearn.preprocessing import scale  
from sklearn.decomposition import PCA  
#add more python packages, as you need to do the work
```

Step Three:

Use Python in Google Colab to download your dataset "b5.csv", to explore and summarize the data as follows:

- Implement the Principal Component Analysis (PCA) for this dataset using `pca = PCA(n_components=2)`.
- Draw a graph of the two components `pca1` and `pc2`.

Rubric

Refer to your Canvas assignment for the rubric.

Submission

After you complete each assignment, you will need to share your completed work by uploading an 'ipynp' and 'PDF' file of your work for your assignment submission. Follow the directions as outlined in the [Colab Setup Instructions](#) and follow the sharing section.

1. Name your Colab file: "YourLastName_FirstName_LAB3.ipynp"
2. Show all work as a PDF: "YourLastName_FirstName_LAB3.pdf"